EXPERIMENT - 4

AIM: Using the Employee schema, write queries with INNER JOIN, LEFT JOIN, selfjoin, 3-way join, correlated subqueries, EXISTS, and simulated INTERSECT and EXCEPT. Compare execution plans using EXPLAIN.

SOURCE CODE:

```
CREATE TABLE Department (
   dept_id INT PRIMARY KEY,
   dept_name VARCHAR(50)
);

CREATE TABLE Project (
   project_id INT PRIMARY KEY,
   project_name VARCHAR(50),
   dept_id INT,
   FOREIGN KEY (dept_id) REFERENCES Department(dept_id)
);

CREATE TABLE Employee (
   emp_id INT PRIMARY KEY,
   emp_name VARCHAR(50),
   dept_id INT,
   project_id INT,
   salary DECIMAL(10,2),
   city VARCHAR(50),
   FOREIGN KEY (dept_id) REFERENCES Department(dept_id),
   FOREIGN KEY (project_id) REFERENCES Project(project_id)
);

INSERT INTO Department VALUES
(1, 'HR'), (2, 'Finance'), (3, 'IT'), (4, 'Marketing'), (5, 'Operations');

INSERT INTO Project VALUES
(101, 'Recruitment System', 1), (102, 'Payroll Automation', 2),
(103, 'Cybersecurity Upgrade', 3), (104, 'Website Revamp', 3),
(105, 'Ad Campaign', 4), (106, 'Market Research', 4),
(107, 'Logistics Optimization', 5), (108, 'Inventory System', 5);

INSERT INTO Employee VALUES
(1, 'Alice', 1, 101, 45000, 'Delhi'),
(2, 'Bob', 1, 101, 42000, 'Mumbai'),
(3, 'Charlie', 1, 101, 47000, 'Pune'),
(4, 'David', 2, 102, 55000, 'Delhi'),
(5, 'Eva', 2, 102, 60000, 'Lucknow'),
(6, 'Frank', 2, 102, 52000, 'Chennai'),
(7, 'Grace', 3, 103, 70000, 'Bangalore'),
(8, 'Hannah', 3, 103, 68000, 'Hyderabad'),
(9, 'Ian', 3, 104, 72000, 'Delhi'),
(10, 'Jack', 3, 104, 65000, 'Pune'),
```

```
(11, 'Karen', 3, 104, 64000, 'Mumbai'),
(12, 'Leo', 4, 105, 48000, 'Delhi'),
(13, 'Mia', 4, 105, 50000, 'Lucknow'),
(14, 'Nina', 4, 106, 51000, 'Chennai'),
(15, 'Oscar', 4, 106, 53000, 'Delhi'),
(16, 'Paul', 5, 107, 56000, 'Hyderabad'),
(17, 'Quinn', 5, 107, 57000, 'Bangalore'),
(18, 'Rita', 5, 107, 59000, 'Delhi'),
(19, 'Sam', 5, 108, 60000, 'Lucknow'),
(20, 'Tina', 5, 108, 61000, 'Mumbai'),
(21, 'Uma', 5, 108, 62000, 'Chennai'),
(22, 'Victor', 2, 102, 53000, 'Delhi'),
(23, 'Wendy', 2, 102, 54000, 'Hyderabad'),
(24, 'Xavier', 3, 103, 71000, 'Bangalore'),
(25, 'Yara', 3, 104, 66000, 'Lucknow'),
(26, 'Zane', 4, 105, 49000, 'Delhi'),
(27, 'Aditi', 4, 106, 52000, 'Mumbai'),
(28, 'Bhavya', 1, 101, 46000, 'Chennai'),
(29, 'Chetan', 1, 101, 43000, 'Lucknow'),
(30, 'Devika', 5, 107, 58000, 'Delhi');

SELECT E.emp_id, E.emp_name, D.dept_name
FROM Employee E
INNER JOIN Department D
   ON E.dept_id = D.dept_id;

SELECT D.dept_id, D.dept_name, E.emp_name
FROM Department D
LEFT JOIN Employee E
   ON D.dept_id = E.dept_id;

SELECT E1.emp_name AS Employee1, E2.emp_name AS Employee2, E1.dept_id
FROM Employee E1
INNER JOIN Employee E2
   ON E1.dept_id = E2.dept_id
   AND E1.emp_id < E2.emp_id;

SELECT E.emp_name, P.project_name, D.dept_name
FROM Employee E
INNER JOIN Project P
   ON E.project_id = P.project_id
INNER JOIN Department D
   ON E.dept_id = D.dept_id;

SELECT E.emp_name, E.dept_id, E.salary
FROM Employee E
WHERE E.salary > (
   SELECT AVG(E2.salary)
   FROM Employee E2
   WHERE E2.dept_id = E.dept_id );
```

```
SELECT E.emp_id, E.emp_name, E.dept_id
FROM Employee E
WHERE EXISTS (
   SELECT 1
   FROM Project P
   WHERE P.dept_id = E.dept_id );

SELECT emp_name
FROM Employee
WHERE emp_id IN (
   SELECT emp_id
   FROM Employee
   WHERE city = 'Delhi' )
AND emp_id IN (
   SELECT emp_id
   FROM Employee
   WHERE salary > 60000 );

SELECT emp_name
FROM Employee
WHERE emp_id NOT IN (
   SELECT emp_id
   FROM Employee
   WHERE city = 'Delhi' );

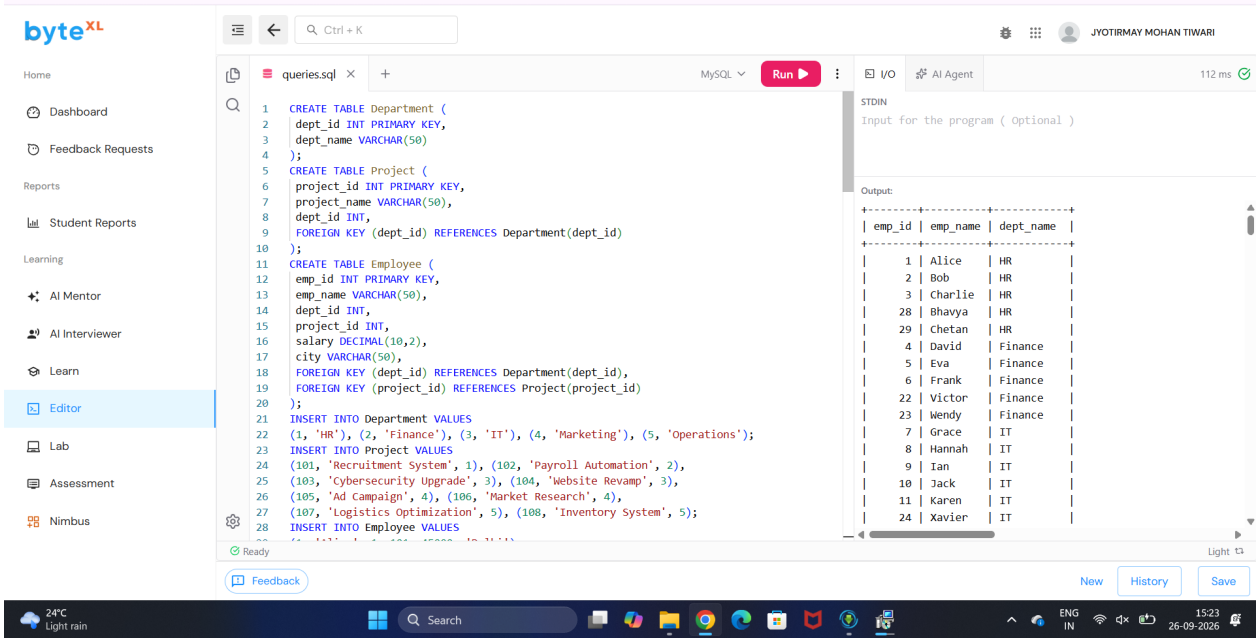
EXPLAIN
SELECT E.emp_name, D.dept_name
FROM Employee E
INNER JOIN Department D
   ON E.dept_id = D.dept_id;

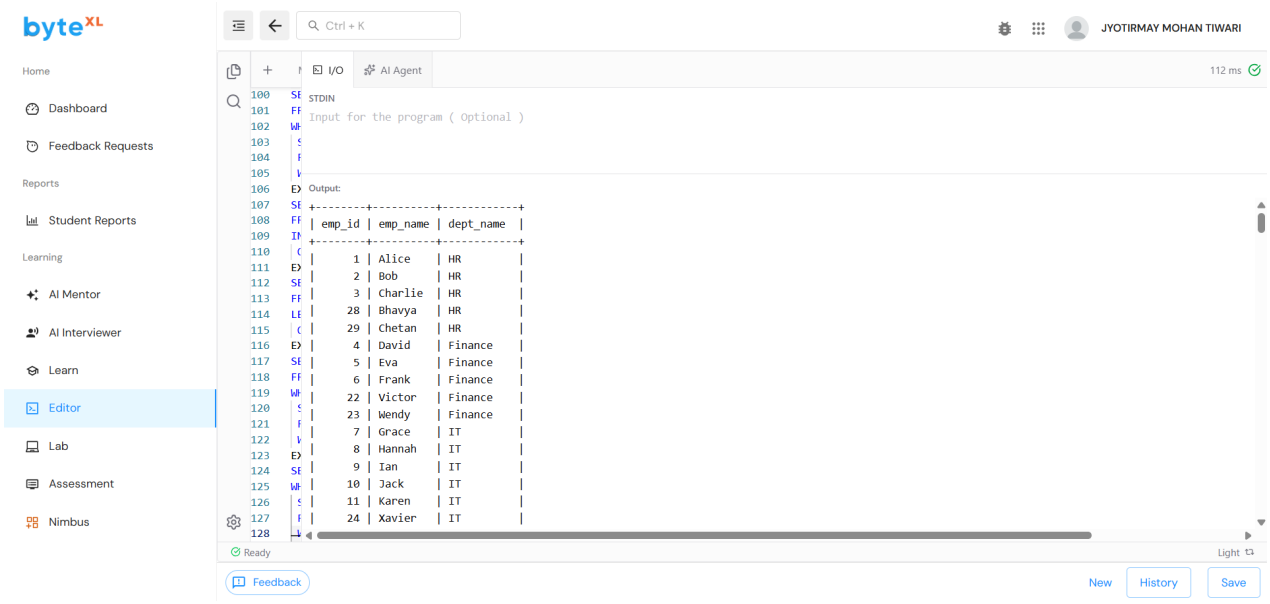
EXPLAIN
SELECT D.dept_name, E.emp_name
FROM Department D
LEFT JOIN Employee E
   ON D.dept_id = E.dept_id;

EXPLAIN
SELECT E.emp_name, E.salary
FROM Employee E
WHERE E.salary > (
   SELECT AVG(E2.salary)
   FROM Employee E2
   WHERE E2.dept_id = E.dept_id );

EXPLAIN
SELECT E.emp_name FROM Employee E
WHERE EXISTS (
   SELECT 1
   FROM Project P
   WHERE P.dept_id = E.dept_id );
```

OUTPUT :-


byteXL
Ctrl + K
JYOTIRMAY MOHAN TIWARI
Home
Dashboard
Feedback Requests
Reports
Student Reports
Learning
AI Mentor
AI Interviewer
Learn
Editor
Lab
Assessment
Nimbus
queries.sql
MySQL
Run
I/O
AI Agent
112 ms
STDIN
Input for the program (Optional)
Output:
CREATE TABLE Department (
dept_id INT PRIMARY KEY,
dept_name VARCHAR(50)
);
CREATE TABLE Project (
project_id INT PRIMARY KEY,
project_name VARCHAR(50),
dept_id INT,
FOREIGN KEY (dept_id) REFERENCES Department(dept_id)
);
CREATE TABLE Employee (
emp_id INT PRIMARY KEY,
emp_name VARCHAR(50),
dept_id INT,
project_id INT,
salary DECIMAL(10,2),
city VARCHAR(50),
FOREIGN KEY (dept_id) REFERENCES Department(dept_id),
FOREIGN KEY (project_id) REFERENCES Project(project_id)
);
INSERT INTO Department VALUES
(1, 'HR'), (2, 'Finance'), (3, 'IT'), (4, 'Marketing'), (5, 'Operations');
INSERT INTO Project VALUES
(101, 'Recruitment System', 1), (102, 'Payroll Automation', 2),
(103, 'Cybersecurity Upgrade', 3), (104, 'Website Revamp', 3),
(105, 'Ad Campaign', 4), (106, 'Market Research', 4),
(107, 'Logistics Optimization', 5), (108, 'Inventory System', 5);
INSERT INTO Employee VALUES
emp_id	emp_name	dept_name
1	Alice	HR
2	Bob	HR
3	Charlie	HR
28	Bhavya	HR
29	Chetan	HR
4	David	Finance
5	Eva	Finance
6	Frank	Finance
22	Victor	Finance
23	Wendy	Finance
7	Grace	IT
8	Hannah	IT
9	Ian	IT
10	Jack	IT
11	Karen	IT
24	Xavier	IT
Ready
Light
Feedback
New
History
Save
24°C
Light rain
Search
ENG
IN
15:23
26-09-2026


byteXL
Ctrl + K
JYOTIRMAY MOHAN TIWARI
Home
Dashboard
Feedback Requests
Reports
Student Reports
Learning
AI Mentor
AI Interviewer
Learn
Editor
Lab
Assessment
Nimbus
I/O
AI Agent
112 ms
STDIN
Input for the program (Optional)
Output:
emp_id	emp_name	dept_name
1	Alice	HR
2	Bob	HR
3	Charlie	HR
28	Bhavya	HR
29	Chetan	HR
4	David	Finance
5	Eva	Finance
6	Frank	Finance
22	Victor	Finance
23	Wendy	Finance
7	Grace	IT
8	Hannah	IT
9	Ian	IT
10	Jack	IT
11	Karen	IT
24	Xavier	IT
Ready
Light
Feedback
New
History
Save